INCH-POUND

MIL-PRF-53027B(AT)
14 March 1996
SUPERSEDING
MIL-H-53027A(ME)
29 May 1985

PERFORMANCE SPECIFICATION

HOSE ASSEMBLY, RUBBER: LIGHTWEIGHT COLLAPSIBLE, 6-INCH; FOR DRINKING (POTABLE) WATER

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 6-inch (in.), lightweight, collapsible rubber hose assembly for drinking (potable) water (see 6.1).

1.2 Classification. The length of the hose assembly is as specified ± 1 percent (%), including the two end couplings (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4720

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-C-10387 - Couplings, Clamp, Pipe; with Bolts and Synthetic - Rubber Gaskets, for Grooved - End Pipe and Tube.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

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PUBLICATIONS

FOOD AND DRUG ADMINISTRATION (FDA)

Title 21, Code of Federal Regulations, Chapter 1, Part 177.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

NATIONAL SANITATION FOUNDATION (NSF)

NSF Standard 14 - Official Plastics Listing.

(Application for copies should be addressed to National Sanitation Foundation, 3475 Plymouth Road, Ann Arbor, MI 48105.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN PUBLIC HEALTH ASSOCIATION, INC. (APHA)

Standard Methods for the Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 18th Street, Washington, DC 20036.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D380	- Standard Test Methods for Rubber Hose (DoD Adopted).
ASTM D412	- Standard Test Methods for Volcanized Rubber and Thermoplastic Rubber and Thermoplastic Elastomers-Tension (DoD Adopted).
ASTM D471	- Standard Test method for Rubber Property-Effect of Liquids (DoD Adopted).
ASTM D518	- Standard Test method for Rubber Deterioration-Surface Cracking (DoD Adopted).
ASTM D1149	- Standard Test Method for Rubber Deterioration-Surface Ozone Cracking in a Chamber (Flat Specimen) (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J 1086	- Numbering Metals and Alloys (DoD Adopted).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.1 Sample selection plan. As part of the first article test, the contractor shall furnish a sample selection plan within the time specified (see 6.2). The plan shall assure all production hose assemblies are processed in the same manner and with production personnel as the first article hose assembly, using the same materials and methods as the first article hose assembly.

3.2 Process data. Within the time span specified (see 6.2), the contractor shall make available to the Government, on the site of manufacture, a description of materials and a documented inspection instruction which will be used by the contractor to assure that each production hose assembly is a duplicate of the first article hose assembly. Change of materials,

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procedures, or sources of supply from those initially selected shall be subject to the approval of the contracting officer (see 6.3).

3.3 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for hoses. Materials selected shall be subject to all provisions of this specification (see 4.4.1 and 4.4.2).

3.3.1 Material deterioration prevention and control. The hose assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the hose assembly may be exposed (see 4.4.1 and 4.4.2).

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion (see 4.4.2).

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative (see 4.4.1 and 4.4.2).

3.3.2 Elastomer or rubber.

3.3.2.1 Tube. The basic material for the tube shall be a natural or a synthetic compound conforming to the Federal Food, Drug and Cosmetic Food Additives, Title 21, Part 177.2600, Chapter 1, Code of Federal Regulations, or the liner material and all surfaces that contact drinking water shall be tested and listed as acceptable for contact with drinking water by the National Sanitation Foundation (NSF) under standard 14. The compounds listed in sections 177.1020, 177.1030, 177.1040, 177.1050, and 177.1480 of the above document shall not be acceptable materials. Questions pertinent to this effect shall be referred by the procuring activity to the departmental medical services, who will act as an advisor to the procuring activity (see 4.4.1).

3.3.2.2 Cover. The cover shall be elastomeric or rubber compound for water, ozone, and oil resistance. The color shall be a visual match to FED-STD-595 color number 30277 (see 4.4.1).

3.3.2.3 Reinforcement material. The reinforcing material shall consist of one or more plies of high-strength synthetic fabric, evenly and firmly fabricated, free from dirt, knots, lumps, irregularities of twist, or other defects (see 4.4.1).

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3.3.3 Unified numbering system (UNS). Where applicable, metal alloys are identified by UNS in accordance with SAE J 1086 (see 4.4.2).

3.4 Design and construction. Hoses shall conform to the manufacturer's design and construction for the applicable item, except for hose assembly dimensions which shall conform to the applicable drawings. The hose assembly shall consist of two hoses sections with end fittings at each end and a grooved coupling at one end (see 4.4.1 and 4.4.2). The length of the hose assembly shall be as specified (see 6.2). A tolerance in length of ± 1 % will be permitted.

3.4.1 Hose assembly. The hose assembly shall consist of two hoses sections with grooved-end adapters on each end and a grooved end coupling on one end. The length of each section, including the two end couplings, shall be as specified (see 6.2). Two of these sections may be connected together with a grooved coupling (see 3.4.4). The hose assembly shall have no twist when stored. The ends shall be sealed to prevent foreign matter from entering the hose (see 4.4.2).

3.4.2 Hose. The hose shall consist of an inner tube, one or more plies of reinforcement material, and a protective cover. All layers shall be bonded to their adjacent layers so as to produce a unified hose wall. The color of the exterior surface of the hose shall be lusterless with a visual match to FED-STD-595, color chip 30277 (see 4.4.2).

3.4.2.1 Dimensions and weight. The hose shall have an inside diameter of 6 in. +0.19 in. or -0.13 in. The length of each hose section, including the two end couplings, shall be as specified (see 6.2). The weight of the hose shall not exceed 2.75 pounds per linear foot (lbs/lin ft) (see 4.4.2).

3.4.3 Hose to grooved adapter. Each end of each hose section shall be equipped with a one piece adapter with grooved end conforming to MIL-C-10387, figure 1. The clamps shall be non-vivid and non-reflective. Adapters shall be firmly affixed to the hose with not less than three corrosion-resisting clamps (see 4.4.2).

3.4.4 Grooved couplings. The grooved couplings shall conform to MIL-C-10387, class A, size 6-in. Use of Victaulic, style 77A or equal is acceptable (see 4.4.2).

3.5 Physical properties.

3.5.1 Adhesion.

3.5.1.1 Original adhesion. The original adhesion between tube and plies, between the plies, and between the cover and the plies, shall be not less than 10 pounds (lbs) (see 4.4.3.4.3).

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3.5.1.2 Adhesion after immersion in water. After immersion in distilled water for 14 days at 158 degrees Fahrenheit (°F) (70 degrees Celsius) (°C)) ± 3.6°F (± 2°C), the adhesion between the tube and plies, between the plies, and between the cover and the plies shall be not less than 6 lbs (see 4.4.3.4.4).

3.5.1.3 Low temperature flexibility. When tested as specified in 4.4.3.4.5, there shall be no evidence of cracking, lack of flexibility, or any other defects.

3.5.2 Tensile strength. The tensile strength of the tube shall be not less than 1,250 pounds per square inch (psi) and the tensile strength of the cover shall be not less than 1,500 psi. The tensile strength of the tube and cover (after immersion), shall be not less than 80% or less than 1000 psi (whichever is greater) of the tensile strength obtained before immersion (see 4.4.3.3).

3.5.3 Ultimate elongation. The ultimate elongation of the tube and cover shall be not less than 300%, and the elongation after immersion shall be not less than 200% (see 4.4.3.3).

3.5.4 Ozone resistance. The hose cover shall withstand 50 parts of ozone per 100 million parts of air (pphm), maintained at + 104°F (+40°C) for 168 hours without evidence of cracking when observed under 7-power magnification (see 4.4.3.3).

3.5.5 Oil resistance. The hose cover shall have a maximum volume swell of 80% after immersion in ASTM No. 3 oil (see 4.4.3.3).

3.5.6 Hydrostatic pressure. The hose assembly shall withstand an internal hydrostatic pressure of not less than 225 psi without rupture or leak (see 6.4), either in the hose body or around the couplings and without any deformation of the couplings (see 4.4.3.2.1).

3.5.7 Elongation and twist. The hose section, when subjected to an internal hydrostatic pressure of 150 psi, shall not show elongation greater than one percent in length or twist, either right or left, by more than 36 degrees per 100 feet of length (see 4.4.3.2.2).

3.5.8 Burst pressure. The hose shall withstand a burst pressure of not less than 500 psi without rupture or failure (see 4.4.3.4.1).

3.6. Taste and odor. The hose shall not impart taste to a rating level higher than 4 or impart odor to an intensity greater than threshold odor No. 2 to chlorinated (drinking) water when tested in accordance with American Public Health Association Standard Methods for the Examinations of Water and Wastewater (see 4.4.3.3.1).

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3.7 Identification marking. Each hose assembly shall have the following information:

a. Date of manufacturer (quarter and year).
b. Contract or order number.
c. Manufacturer's name and trademark.
d. The words "DRINKING WATER ONLY."

Hoses shall be branded at intervals not greater than 10 feet (ft). The letters shall be not less than 0.18-in. high. Label color shall contrast with hose cover color. Each hose assembly shall be marked with a longitudinal blue stripe not less than 0.125-in. wide and extending the full length of the hose. The label may substitute for the longitudinal stripe, provided it is continuous for the full length of the hose and is blue in color. The label may be within the longitudinal stripe provided the width of the strips is not less than 0.20-in. greater than the height of the label letters, and the label is in contrast to the color of the stripe. The label and strips shall remain clear, distinct and legible for the life of the hose and shall not be adversely affected by handling, bending, water, oil, or environmental effects (see 4.4.2).

3.8 Metallic component finish. Metallic parts shall be coated and colored to approximate color No. 30277 of FED-STD-595 (see 4.4.2).

3.9 Workmanship. The hose assembly shall have no defects affecting appearance or serviceability. The hose shall be of uniform quality and shall be free from cracks, pitting, or other defects (see 4.4.2).

3.10 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the hose assemblies may be newly fabricated from recovered materials to the maximum extent practicable, provided the hose assemblies produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the hose assemblies (see 4.4.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

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4.2 First article inspection.

4.2.1 Examination. The first article hose assembly shall be examined for the defects as specified in table I. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests. The first article hose assembly shall be subjected to the tests marked "X" of table II. Samples for adhesion tests shall be cut from a length of hose. Samples of the synthetic rubber compound shall be representative of materials used to fabricate the hose assembly. Failure of any test shall be cause for rejection of the hose assembly.

4.3 Conformance inspection.

4.3.1 Sampling.

4.3.1.1 Examination. Sampling of hose assemblies for examination shall be in accordance with ANSI/ASQC Z1.4.

4.3.1.2 Tests.

4.3.1.2.1 Elastomeric or rubber compound. Four test slabs, each 6 in. by 6 in. by 0.075 in. $\pm$ 0.010 in. thick, of each elastomeric and natural rubber compound and of equivalent cure used to fabricate the hose assemblies, shall be supplied with each lot.

4.3.1.2.2 Hose. Sampling of hose for tests shall be in accordance with ANSI/ASQC Z1.4. Test sections shall be cut from the hose and shall be of sufficient length for the performance of the required tests.

4.3.2 Examination. Sample hose assemblies selected in accordance with 4.3.1.1 shall be examined as specified in 4.4.2.

4.3.3 Tests.

4.3.3.1 Individual. Each hose assembly shall be subjected to the test marked "X" of table II. Failure of the tests shall be cause for rejection.

4.3.3.2 Samples. Samples selected in accordance with 4.3.1.2.1 and 4.3.1.2.2 shall be subjected to the tests marked "X" of table II. Failure of any test shall be cause for rejection of the hose assemblies represented by the samples.

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4.4 Methods of inspection.

4.4.1 Material design and construction. Conformance to 3.3 through 3.3.2.3 and 3.4 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.4.2 Examination. The hose assembly shall be examined for the defects specified in table I.

TABLE I. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Material not as specified (see 3.3).	Visual
102	Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment (see 3.3.1).	
103	Dissimilar metals are not effectively insulated from each other (see 3.3.1.1).	Visual
104	Contractor does not have documentation available for identification of material; material finishes or treatments (see 3.3.1.2).	Visual
105	Components missing (see 3.4).	Visual
106	Identificationof metal alloys not as specified (see 3.3.3).	Visual
107	Hose assembly not as specified (3.4.1).	Visual/Functional
108	Hose not as specified (see 3.4.2).	Visual
109	Dimensions and weight not as specified (see 3.4.2.1).	Visual/SIE
110	Hose to grooved adapter not as specified (see 3.4.3).	Visual/SIE
111	Bands not as specified (see 3.4.3).	Visual
112	Grooved couplings not as specified (see 3.4.4).	Visual
113	Marking missing, incomplete, or illegible(see 3.7).	Visual
114	Finish not as specified (see 3.8).	Visual
115	Workmanship not as specified (see 3.9).	Visual
116	Used, rebuilt, or remanufactured components pieces, or parts incorporated into the hose assembly (see 3.10).	Visual

SIE = Standard Inspection Equipment.

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4.4.3 Tests.

4.4.3.1 Test schedule. Hose assembly shall be subjected to the tests as specified in table II.

TABLE II. Classification of tests.

Test	Requirement	Test	First article	Conformance	
				Individual	Sample
Hose assembly					
Hydrostatic pressure	3.5.6	4.4.3.2.1	X	X	-
Elongation and twist	3.5.7	4.4.3.2.2	X	X	X 1/
Elastomeric or rubber compounds					
Initial tensile strength	3.5.2	4.4.3.3	X	-	X
Tensile strength after immersion	3.5.2	4.4.3.3	X	-	X
Ultimate elongation, initial	3.5.3	4.4.3.3	X	-	X
Ultimate elongation after immersion	3.5.3	4.4.3.3	X	-	X
Ozone resistance	3.5.4	4.4.3.3	X	-	X
Oil resistance	3.5.5	4.4.3.3	X	-	X
Taste and odor	3.6	4.4.3.3.1	X	-	-
Hose					
Burst pressure	3.5.8	4.4.3.4.1	X	-	X
Weight	3.4.2.1	4.4.3.4.2	X	-	X
Original adhesion	3.5.1.1	4.4.3.4.3	X	-	X
Adhesion after immersion in water	3.5.1.2	4.4.3.4.4	X	-	X
Low temperature flexibility	3.5.1.3	4.4.3.4.5	X	-	X 1/

1/ One lot testing required for each ten lots submitted.

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4.4.3.2 Hose Assembly.

4.4.3.2.1 Hydrostatic pressure. Each hose section shall be subjected to the hydrostatic tests, using water as a test fluid, as specified in ASTM D380 for proof pressure. Nonconformance to 3.5.6 shall constitute failure of this test. The hydrostatic test shall be performed with drinking (potable) water. After testing, the hose shall be dried and the ends shall be sealed.

4.4.3.2.2 Elongation and twist. Elongation and twist measurement shall be on a hose length not less than 100 ft, laid out straight. One end of the hose shall be fixed and not allowed to turn. Twist shall be measured at the free end and recorded as clockwise or counterclockwise. The following procedure shall be completed: Pressurize the hose section, using drinking (potable) water as a test fluid, to 10 psi and hold for not more than two minutes and mark the length and twist position. After two minutes increase the pressure to 150 psi and mark the length and twist position. Nonconformance to 3.5.7 shall constitute failure of this test.

4.4.3.3 Elastomeric or rubber compounds. Elastomeric or rubber compound samples shall be tested as follows:

Property	Test method
Initial tensile strength	ASTM D412
Tensile strength after immersion	ASTM D471 2/
Ozone resistance	ASTM D1149 1/
Volume swell	ASTM D471 3/
Initial ultimate elongation	ASTM D412
Ultimate elongation after immersion	ASTM D471 2/

1/ Specimens shall be mounted in accordance with procedure B, ASTM D518, and ozone concentration shall be 50 ± 10 parts per 100,000,000 (pphm). Time exposure shall be 168 hours, temperature of exposure shall be 104°F (40°C).
2/ The samples shall be immersed in distilled water for 14 days at 158°F (70°C) ± 3.6°F (2°C).
3/ Hose cover samples shall be exposed in ASTM No. 3 oil for 70 hours at 212°F (100°C).

Nonconformance to 3.5.2, 3.5.3, 3.5.4, and 3.5.5 shall constitute failure of these tests.

4.4.3.3.1 Taste and odor. Taste and odor test shall be performed in accordance with procedures outlined in American Public Health Association, Standard Methods for Examination of Water and Wastewater. A sample of the hose shall be filled with distilled water having a 0.2 parts per million of total available chlorine at the start of the 70-hour test. Sample filled with

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water shall be maintained at 180°F (82°C) for 30 minutes and then shall be allowed to cool to 99°F (37°C). The sample shall be maintained at 99°F (37°C) for the remaining 69.5 hours. The samples shall be long enough to expose 5.6 square inch (in^2) (36 square centimeters (cm^2)) of inner tube to one liter of chlorinated distilled water. Nonconformance to 3.6 shall constitute failure of this test.

4.4.3.4 Hose.

4.4.3.4.1 Burst pressure. The test section of hose selected under 4.3.1.2.2 shall be subjected to the straight bursting test as specified in ASTM D380. Nonconformance to 3.5.8 shall constitute failure of this test.

4.4.3.4.2 Weight. A section 12 in. long shall be cut from the test section of hose selected under 4.3.1.2.2 and weighed. Nonconformance to 3.4.2.1 shall constitute failure of this test.

4.4.3.4.3 Original adhesion. From the test section of hose selected under 4.3.1.2.2, ring specimens shall be prepared in accordance with ASTM D380. The adhesion of the ring specimen shall be determined in accordance with ASTM D380, machine method. Nonconformance to 3.5.1.1 shall constitute failure of this test.

4.4.3.4.4 Adhesion after immersion in water. Specimens prepared as described in 4.4.3.4.3 shall be immersed as specified in 3.5.1.2. Within 5 minutes after removal from immersion, the adhesion shall be determined as specified in 4.4.3.4.3. Nonconformance to 3.5.1.2 shall constitute failure of this test.

4.4.3.4.5 Low temperature flexibility. A sample 8-ft length of hose shall be folded 180 degrees and tied in the folded position. The hose shall then be conditioned for 72 hours at - 25°F (- 31.7°C). The hose shall be removed from the test chamber and unfolded 180 degrees using a force of not more than 10 lbs within 5 seconds. The unfolding shall be accomplished within 3 minutes after removal from the test chamber. The hose sample shall then be hydrostatic pressure tested at 225 pounds per square inch gage (psig) in accordance with ASTM D380. Hose fabricated with splices shall have two hose test samples, one hose sample folded without a splice, and the other sample folded at the splice. Nonconformance of any test sample to 3.5.1.3 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory

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Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The hose assembly is intended for use with light-weight, collapsible-hose pipelines in drinking (potable) water systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Specification part number required (see 1.2.).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
d. When a first article is required for inspection and approval and the number of units required (see 3.1).
e. Time frame required for submission of sample selection plan and process data (see 3.1.1 and 3.2).
f. Length of hose assembly required (see 3.4).
g. Length of each hose section required (see 3.4.1 and 3.4.2.1).
h. Packaging requirements (see 5.1).
i. Any special marking instruction to be specified by the contracting officer.

6.3 First article. When a first article inspection is required, the item will be tested and should be an initial production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Definitions.

6.4.1 Leak. A leak is defined as any passage of fluid from the inner portion of the hose assembly to the outer portion of the hose assembly as determined by sight or touch. A leak includes but is not limited to fluid passing through a hose assembly fitting, fluid passing through the hose body, or fluid passing through the junction of the hose and an attached fitting.

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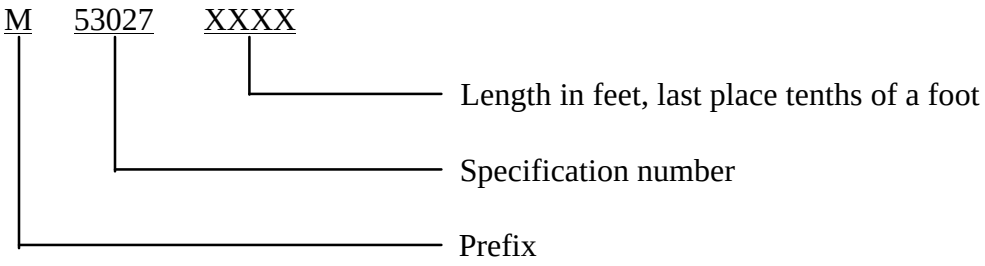
6.4.2 Rupture. A rupture in the hose assembly is defined as a split or crack in the hose, or any split or crack at the junction of hose and a fitting.

6.5 Part or identifying number. The part or identifying number for the hose assemblies as specified (see 6.2) is in feet with the last number expressed in tenths of a foot and should be designated as follows:



EXAMPLE: M53027-5000, Hose Assembly, per MIL-PRF-53027, 500.0 ft long.

6.6 Disposition of test assemblies. Hoses undergoing destructive tests should not be used and should be indelibly marked "DO NOT USE" and purposely scrapped.

6.7 Subject term (key word) listing.

Beverage
Collapsible-hose pipelines
Fittings nonmetallic
Food service, sanitary medical

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 4720-0088)

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